MAT627 Chapter 2.7 Programming Assignment

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1 Finite-Difference Model

For Parts 1–3, the boundary value problem is

$$-u''(x) + u(x) = f(x), \qquad 0 < x < 1, \qquad u(0) = u(1) = 0.$$

On the uniform mesh $x_k = kh$ with $h = 1/N$, the interior unknowns satisfy

$$-\frac{U_{k-1} - 2U_k + U_{k+1}}{h^2} + U_k = f(x_k), \qquad k = 1, \ldots, N-1,$$

or equivalently

$$-U_{k-1} + (2 + h^2)U_k - U_{k+1} = h^2 f(x_k).$$

This produces a tridiagonal linear system, which is solved with Algorithm 2.6 (Thomas elimination and back substitution).

For Part 4, the modified problem is

$$-u''(x) + u'(x) + u(x) = f(x), \qquad 0 < x < 1, \qquad u(0) = u(1) = 0,$$

and the first derivative is approximated by the backward difference

$$u'(x_k) \approx \frac{u(x_k) - u(x_k - h)}{h}.$$

The resulting discrete equation is

$$-\frac{U_{k-1} - 2U_k + U_{k+1}}{h^2} + \frac{U_k - U_{k-1}}{h} + U_k = f(x_k),$$

that is,

$$-(1 + h)U_{k-1} + (2 + h + h^2)U_k - U_{k+1} = h^2 f(x_k).$$

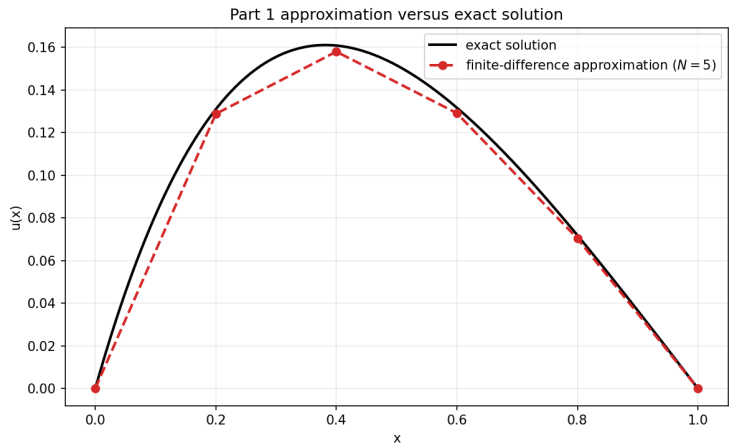

Figure 1: Part 1: nodal finite-difference approximation with $N = 5$ versus the exact solution.

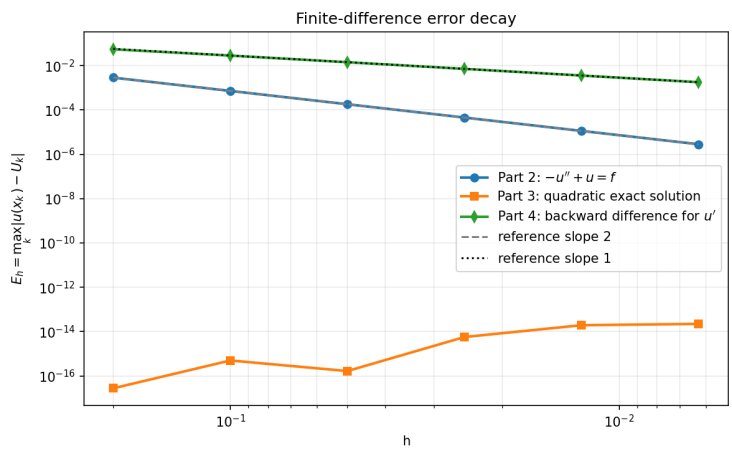

Figure 2: Error decay for the convergence studies in Parts 2–4.

Table 1: Finite-difference solution for $-u'' + u = 4e^{-x} - 4xe^{-x}$ with $N = 5$.

x_k	U_k	$u(x_k)$	$u(x_k) - U_k$
0.2	1.288292×10^{-1}	1.309969×10^{-1}	2.167680×10^{-3}
0.4	1.580141×10^{-1}	1.608768×10^{-1}	2.862697×10^{-3}
0.6	1.291688×10^{-1}	1.317148×10^{-1}	2.545964×10^{-3}
0.8	7.036635×10^{-2}	7.189263×10^{-2}	1.526284×10^{-3}

2 Part 1

3 Part 2

Discrete error for $-u'' + u = 4e^{-x} - 4xe^{-x}$ with exact solution $u(x) = x(1-x)e^{-x}$.

Table 2: Maximum nodal error for part 2.

N	h	$E_h = \max_k \lvert u(x_k) - U_k \rvert$	Rate
5	0.2	2.862697×10^{-3}	–
10	0.1	7.199515×10^{-4}	1.991403
20	0.05	1.803450×10^{-4}	1.997140
40	0.025	4.520950×10^{-5}	1.996062
80	0.0125	1.130343×10^{-5}	1.999865
160	0.00625	2.825924×10^{-6}	1.999966

4 Part 3

Discrete error for $-u'' + u = 2 + x - x^2$ with exact solution $u(x) = x(1-x)$.

Table 3: Maximum nodal error for part 3.

N	h	$E_h = \max_k \lvert u(x_k) - U_k \rvert$	Rate
5	0.2	2.775558×10^{-17}	–
10	0.1	4.996004×10^{-16}	–
20	0.05	1.665335×10^{-16}	–
40	0.025	5.717649×10^{-15}	–
80	0.0125	1.945666×10^{-14}	–
160	0.00625	2.231548×10^{-14}	–

5 Part 4

Discrete error for $-u'' + u' + u = -(x-1)(x^2-11)$ using the backward difference $(u(x) - u(x-h))/h$ for u'.

Table 4: Maximum nodal error for part 4.

N	h	$E_h = \max_k \lvert u(x_k) - U_k \rvert$	Rate
5	0.2	5.445658×10^{-2}	–
10	0.1	2.828373×10^{-2}	0.945134
20	0.05	1.418957×10^{-2}	0.995142
40	0.025	7.106748×10^{-3}	0.997569
80	0.0125	3.558012×10^{-3}	0.998118
160	0.00625	1.779658×10^{-3}	0.999471

6 Discussion

In Part 2, the observed rate approaches 2, which matches the centered second-difference truncation error for $-u'' + u = f$.

In Part 3, the exact solution is the quadratic polynomial

$$u(x) = x(1-x) = x - x^2.$$

The centered second-difference formula is exact for quadratics, so the grid values satisfy the discrete equations up to roundoff. That is why the errors in the table are essentially machine precision and the rate column is omitted.

In Part 4, the second derivative is still approximated with a second-order centered difference, but the backward approximation for u' is only first-order accurate. Consequently, the overall scheme becomes first-order, and the observed rate approaches 1.